

Step 25– Shipping Cost Analysis

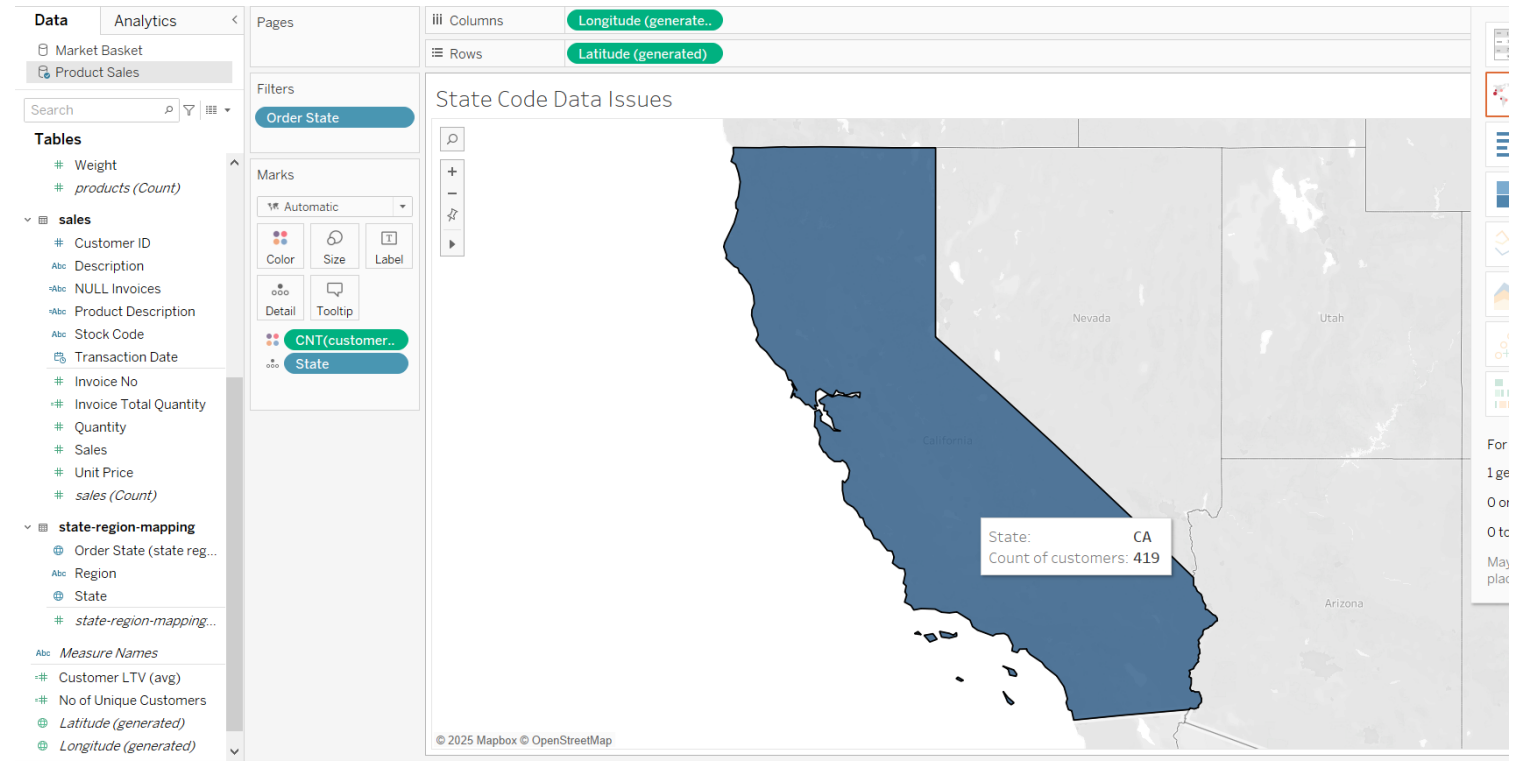
Step 25– Shipping Cost Analysis

- Tableau recognizes geographical elements such as country and state automatically.
- There are limitations if your geographic attributes have multiple variations of values.
- We will look at a specific example of this limitation and devise a solution using a custom state mapping in the data source.
- You will be working with the Product Sales data source.
- **Add a new sheet, "State Code Data Issues".**
- Create a text table showing the number of customers by Order State.
- Filter Order State and keep all variations of the state name "California" or state code "CA".
- Make a note of the number of customers in "CA" and the total number of customers for all different variations of state code "CA" or state name "California".
- Change the text table to a map.
- Tableau only recognized the state code **CA**, leaving the other six records unmapped.
- You can fix this by adding a custom state mapping.
- Add **state-region-mapping.csv** from the *Datasources* folder to the **Product Sales** data source and rename the CSV table to "**state-region-mapping**" and join on the state in the **Customers** table.
- Replacing **Order State** with **State** from the custom mapping will cause unmapped variations of California to map correctly to the state. "Unknown" and "AE" locations refer to overseas US military so ignore it.

Step 26– Shipping Cost Analysis

State Code Data Issues

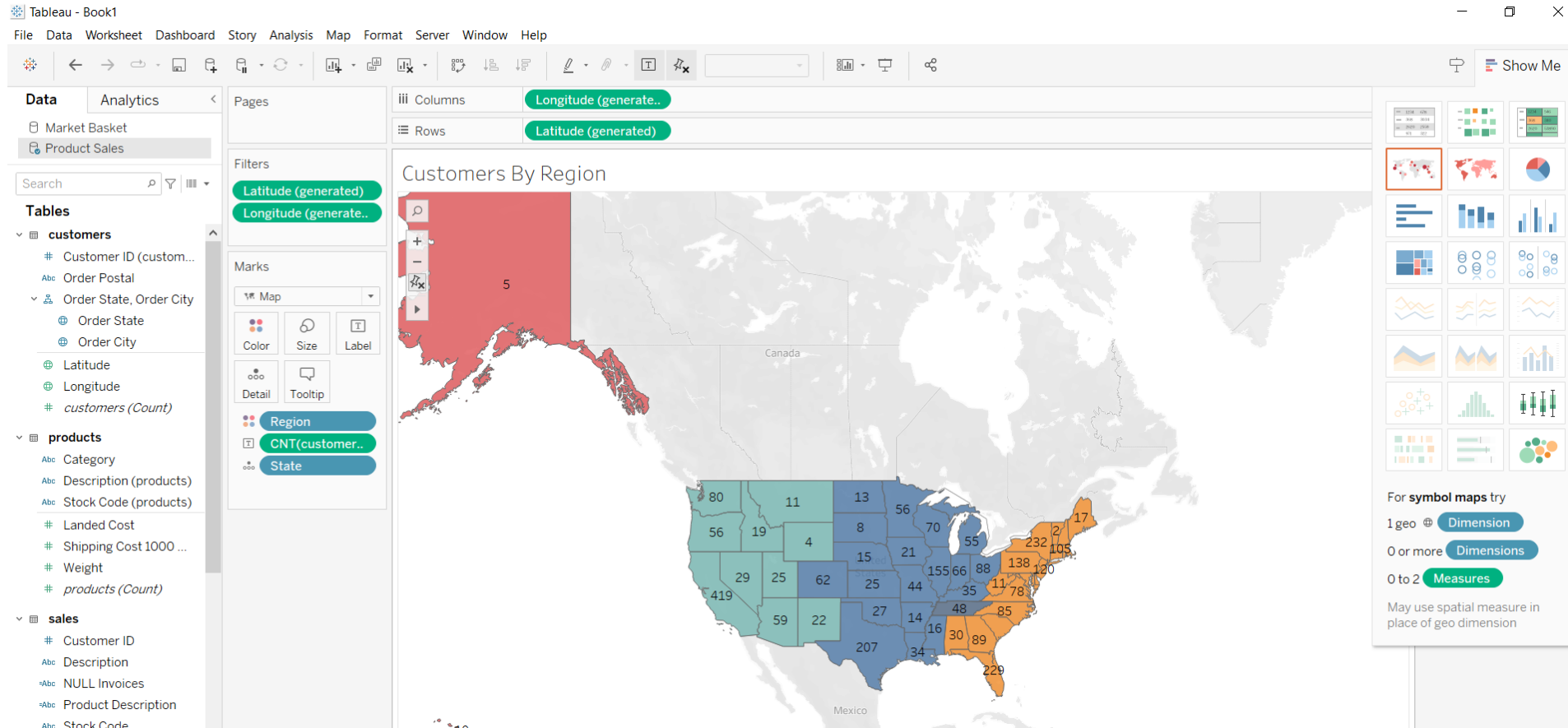
Order State	
CA	368
ca	7
Ca	7
Ca.	3
california	2
CALIFORNIA	20
California	12



Step 27– Shipping Cost Analysis

- Duplicate the sheet **Customers by State** created previously and rename it "**Customers by Region**".
- Move this new sheet to the farthest right in the workbook.
- Replace **Order State** with **State** and add color by **Region** and **Number of Customers** as a data label.
- **How many additional records are associated with the state of California after you added the custom region mapping? If your answer is 51 please proceed.**

Step 27– Shipping Cost Analysis



Step 28– Shipping Cost Analysis

- Management wants to understand where its customers are and the total sales by state.
- To build this map, you will use spatial functions and build a **distance** metric.
- The **makepoint()** function creates geometric points on earth while **distance()** calculates the distance between two locations on earth.
- The store has a warehouse in Los Angeles, CA. The latitude and longitude are **(34.0544, -118.2439)**. Using Tableau's spatial functions, create a spatial object or a geographic point, "**Warehouse-LAX**".
- Make a calculated field "Warehouse-LAX" as the following.

Warehouse-LAX

Product Sales

```
MAKEPOINT(34.0544, -118.2439)|
```

- Every customer address includes latitude and longitude. This is the destination for each order.
- Create a spatial field "**Destination**" using the customer's latitude and longitude.
- MAKEPOINT([Customer Latitude], [Customer Longitude])

Destination

Product Sales

```
MAKEPOINT([Latitude], [Longitude])
```

|

Step 29– Shipping Cost Analysis

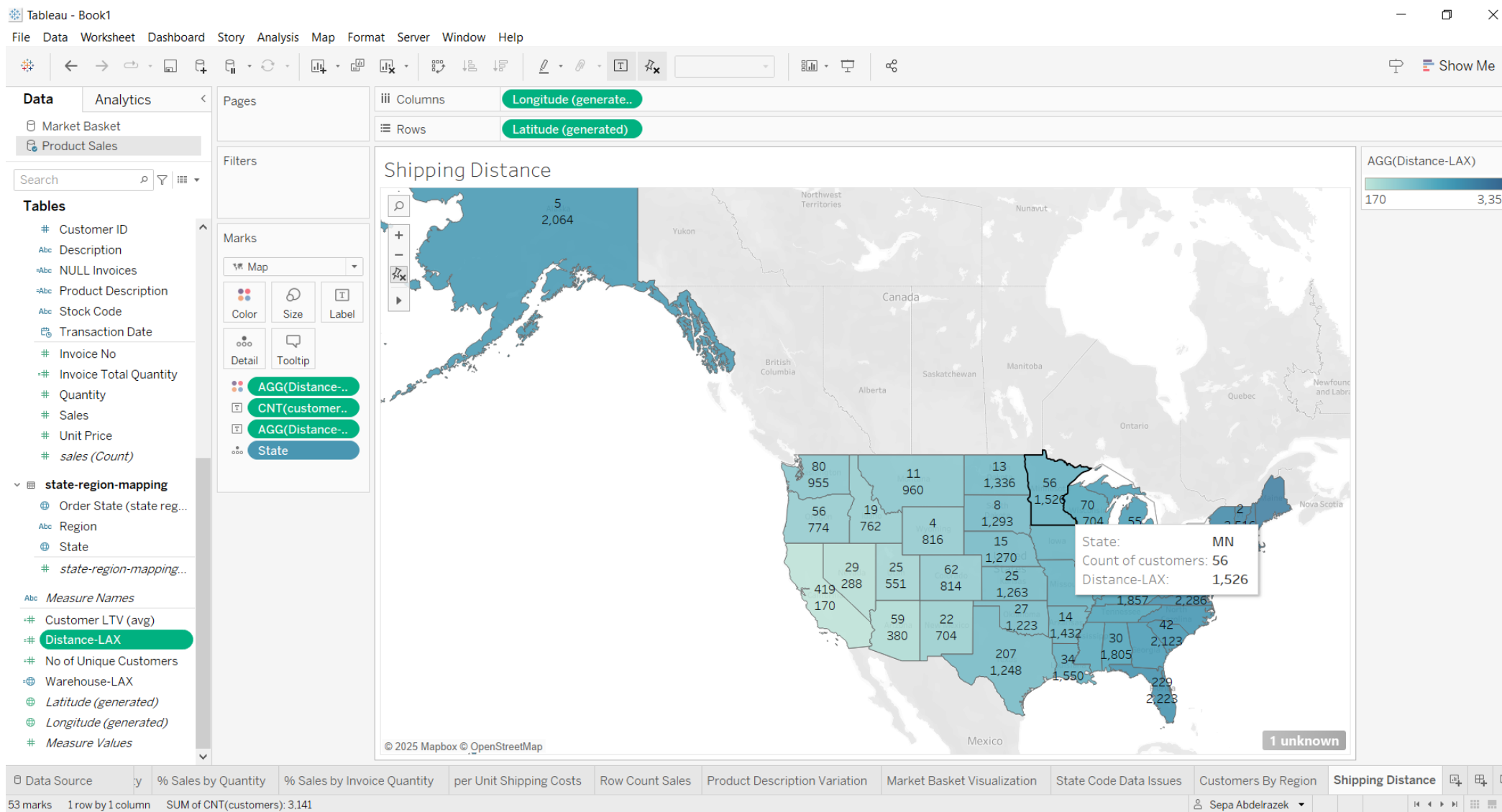
- You now have two geographic or spatial fields defined.
- Calculate the distance between these two points as "**Distance-LAX**".
- Use average aggregation and miles as the unit.

Distance-LAX  Product Sales

```
avg(DISTANCE([Warehouse-LAX], [Destination], "mi"))|
```

- Add a new sheet, "**Shipping Distance**", and create a map of the US and color states by the average distance to the LAX warehouse.
- Add the number of customers and the distance to the tooltip.
- **What is the average shipping distance in miles between New York and the Los Angeles (LAX) warehouse?**
- If your answer is 2426 then proceed

Step 29– Shipping Cost Analysis



Step 30– Shipping Cost Analysis

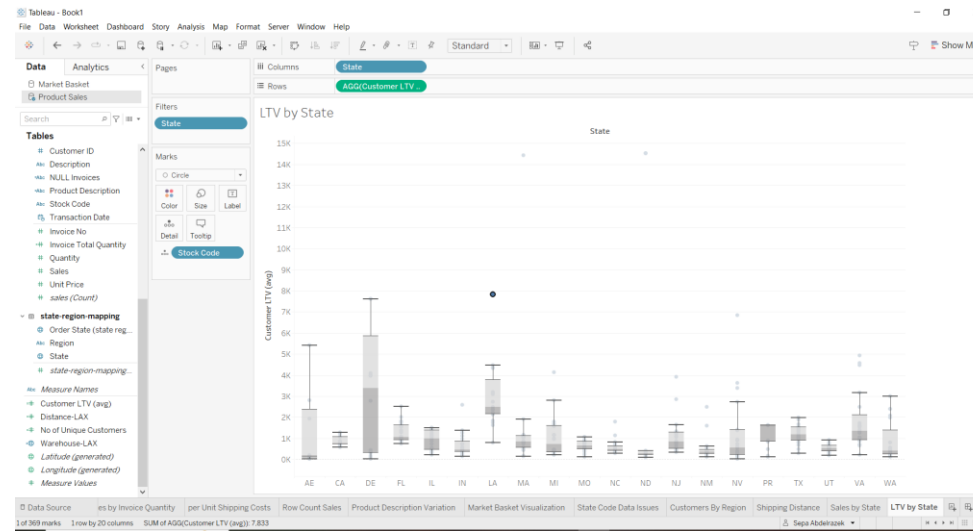
- Create a new sheet, "Sales by State", and add a text table using State and Region dimensions from the state-region-mapping table.
- Add the Sales measure to the table.

The screenshot shows the Tableau interface with a text table titled "Sales by State". The table is organized by Region (Central, East) and lists various states with their corresponding sales values. The interface includes a "Rows" shelf with "Region" and "State", and a "Marks" shelf with "SUM(Sales)".

Region	State	Sales
Null	Null	325,798
Central	AR	4,261
Central	CO	16,630
Central	IA	6,833
Central	IL	60,923
Central	IN	25,223
Central	KS	4,810
Central	KY	9,909
Central	LA	23,761
Central	MI	20,178
Central	MN	15,811
Central	MO	14,664
Central	MS	2,376
Central	ND	16,611
Central	NE	4,371
Central	OH	29,155
Central	OK	7,137
Central	SD	2,253
Central	TN	12,512
Central	TX	97,361
Central	WI	20,203
East	AL	6,108
East	CT	9,832
East	DC	2,230
East	DE	9,139
East	FL	107,523
East	GA	28,128

Step 31– Shipping Cost Analysis

- We created a box-and-whisker plot of average **Customer LTV** by **Order State**.
- We will recreate this chart using the **State** field from **state-region-mapping** and see the impact of using the custom state mapping.
- Duplicate LTV by Order State (created in Before) as "LTV by State" and replace the old state fields with the new fields from the custom mapping.
- Move the duplicated and renamed sheet to the furthestmost right. Remove the top 20 filter on Order State and replace it with the top 20 filter on State.
- **What is the median customer lifetime value in the state code VA? If 1366 proceed.**
- Note that, The custom mapping increased the record count in Virginia by **331**. This improved the accuracy of the median value from an earlier value of **5,772** and reduced the gap with the overall customer LTV of **1,366** across all states.



Step 32– Shipping Cost Analysis

- Due to a lack of automatic shipping cost capture at the transaction level, the shipping department relies on spreadsheets.
- The shipping department has told you that shipping more than one quantity of an item costs, on average, **70% of the cost of a single-unit shipment**. There are variances based on the product.
- You want to allow the users to play around with effective shipping rates as the quantity changes.
- Create a "**Shipping (Baseline)**" field. If the quantity is just a single item, then there is a regular shipping cost (**Shipping Cost 1000 mile**). If more than one item is shipped, the cost per unit shipped is **70% for an additional quantity above one**. For example, if five items are shipped, then the shipping is:

100% of the cost of the first unit

+

$0.7 * ((5-1) * \text{the cost of unit shipment})$

Step 32– Shipping Cost Analysis

Shipping (Baseline)

Product Sales

```
if [Quantity] = 1 then [Shipping Cost 1000 mile]
ELSE
[Shipping Cost 1000 mile] + (([Quantity]-1 )* [Shipping Cost 1000 mile] * 0.7 )
END
```

Step 33– Shipping Cost Analysis

- **Create a parameter** What-if quantity of integer type and allow values from 1-20 in steps of one.
- Set the current value to **five**.
- **Create a calculated field**, Blended Shipping Cost Factor, which calculates the discounted shipping cost based on the following table.
- The parameter allows the user to try different values for effective shipping rates rather than using a fixed value of 0.7 used in step one.
- The formula below maps a set of parameter values to a fixed cost multiplier.

‘What-if quantity’

<= (less than or equal), cost multiplier (output)

1, 1

2, 0.8

4, 0.6

7, 0.5

9, 0.4

else, 0.3

Step 34– Shipping Cost Analysis

- Duplicate the **Shipping (Baseline)** field and rename it to "**Shipping (What-if)**".
- Use **Blended Shipping Cost Factor** in place of the hard-coded 0.7 value.
- Define the field "**Shipping (Difference Baseline)**" as the difference between **Shipping (Baseline)** and **Shipping (What-if)**.
- On a new sheet, "**Shipping Metrics**", add the three shipping metrics to a text table.
- **What is the value of Shipping(What-if)? If 327,037 proceed**
- **Note that Total Shipping Cost decreased to 327,037 by when the avg shipped quantity is 5.**

Shipping Metrics

Shipping (Baseline)	386,525
Shipping (Difference Bas..	59,488
Shipping (What-if)	327,037